

Task 1

Prepared By Eng. Mariam Badr

Q1. what is the value of x, y, z and w (solve it in text file or using comments in dart file)

int x=3;

int y=2;

int z=x++;

int w=++y;

Q2 Compound Interest Calculator (Data Types + Operators)

Calculate the compound interest using the following formula:

A = P * (1 + r/100) * n

```
input:
Enter principal amount: 1000
Enter interest rate (%): 5
Enter years: 3

output:
Final Amount = 1157.625
Interest Earned = 157.625
```

Q3. Research Question - How to Read User Input in Dart?

Search and explore the different ways to read data from the user in Dart

Q4. Distance & Speed Calculator (Arithmetic Operators + Data Types)

Calculate the time needed to travel a certain distance at a given speed, then convert the time from hours to minutes a and seconds.

```
input:
Enter distance (km): 150
Enter speed (km/h): 60

output:
Time = 2.5 hours
Time = 150.0 minutes
Time = 9000.0 seconds
```

Q5. (Bonus+)

(Compute the volume of a cylinder) Write a program that reads in the radius and length of a cylinder and computes the area and volume using the following formulas:

area = radius * radius * pi

volume = area * length

for example

input

```
Enter the radius and length of a cylinder:
5.5
12
```

output

```
The area is 95.0331

The volume is 1140.4
```